BLOOD BANK SYSTEM DEVELOPMENT ROADMAP

Sequential Implementation Order & Dependencies for the 25 Group Issues

Use this timeline to coordinate backend database dependencies before frontend component bindings.

Phase 1: Foundation (Database & Base Configs)

Establish schemas, security wrappers, and design systems first. No UI components or controllers can be tested without this foundation.

[BE-01]	**PostgreSQL Database Schema** Assignee: Bekam	Prerequisite(s): None (Starting Point)
[BE-02]	**Private Laboratory Database** Assignee: Dagim	Prerequisite(s): None (Starting Point)
[BE-04]	**Dual Prisma Client Configuration** Assignee: Bekam	Prerequisite(s): BE-01, BE-02
[BE-03]	**Development Database Seeder** Assignee: Daniel	Prerequisite(s): BE-04
[BE-05]	**JWT Authentication Middleware** Assignee: Dagim	Prerequisite(s): BE-01
[FE-14]	**Global Design System (CSS)** Assignee: Desta	Prerequisite(s): None
[FE-13]	**Central API Service** Assignee: Ebisa	Prerequisite(s): FE-14
[FE-01]	**Application Router** Assignee: Ebisa	Prerequisite(s): FE-13

Phase 2: Core Workflows (Authentication & Stations)

Implement user login, verification screens, donor registration wizard, and sample collection workflows.

[BE-06] **Authentication & Registration Controller**
Assignee: Daniel
Prerequisite(s): BE-05, BE-03

[FE-02] **Login Page UI**
Assignee: Desta
Prerequisite(s): FE-01, BE-06

[FE-03] **Registration Wizard UI**
Assignee: Desalegn
Prerequisite(s): FE-01, BE-06

[BE-07] **Donation Station Controller**
Assignee: Bekam
Prerequisite(s): BE-06

[FE-04] **Station Dashboard UI**
Assignee: Ebisa
Prerequisite(s): FE-13, BE-07

Phase 3: Laboratory & Donor Portals

Verify blood diagnostics in the lab and build the gamified donor portal.

[BE-08] **Laboratory Controller**
Assignee: Dagim
Prerequisite(s): BE-07, BE-04

[FE-05] **Laboratory Dashboard UI**
Assignee: Desta
Prerequisite(s): FE-13, BE-08

[BE-10] **Donor Controller**
Assignee: Bekam
Prerequisite(s): BE-08

[FE-07] **Donor Dashboard UI**
Assignee: Ebisa
Prerequisite(s): FE-13, BE-10

[FE-11] **Announcements Component**
Assignee: Desta
Prerequisite(s): FE-07

[FE-12] **Donor Leaderboard**
Assignee: Desalegn
Prerequisite(s): FE-07

Phase 4: HMS & Administration (Hospital / Admin / Warehouse)

Integrate hospital operations, warehouse inventory checks, and final administrative stats.

[BE-09] **Hospital & Inter-Hospital (H2H) Controller**
Assignee: Daniel
Prerequisite(s): BE-01

[FE-06] **Hospital Dashboard UI**
Assignee: Desalegn
Prerequisite(s): FE-13, BE-09

[FE-09] **Warehouse Dashboard UI**
Assignee: Desalegn
Prerequisite(s): FE-06

[BE-11] **Admin Controller**
Assignee: Dagim
Prerequisite(s): BE-01

[FE-08] **Admin Dashboard UI**
Assignee: Desta
Prerequisite(s): FE-13, BE-11

[FE-10] **Custom SVG Analytics**
Assignee: Ebisa
Prerequisite(s): FE-08